this relative performance is amplified when the augmentation percentage is larger. The behavior that we observe is consistent with our theoretical results in Section 4, as well as our human motion experiments in Section 7.

7. Human Motion Synthesis

Theorem 4.3 states that, in theory, iterative fine-tuning with correction should be more stable than iterative fine-tuning without correction. Crucially, the stability estimates that we prove rely on the dataset size, the synthetic augmentation percentage, how expressible the generative model class is, and having an idealized correction function. To validate how our theory works beyond toy examples, we conduct a case study on human motion synthesis with diffusion models (Tevet et al., 2023). We believe this is a natural setting to test our iterative fine-tuning with correction framework, because synthesizing natural motions is a challenging problem, but there is a natural and intuitive way to automatically correct them at scale–namely, using a physics simulator.

7.1. Generative Model

For our generative model, we use the Human Motion Diffusion Model (MDM) (Tevet et al., 2023). This is a classifier-free diffusion-based generative model for the text-to-motion generation task, where the model receives as input a description of a motion sequence (e.g. "get down on all fours and crawl across the floor"), and outputs a sequence of skeleton poses which attempt to embody that prompt. Synthesizing human motion is challenging not only for the diverse and compositional text prompts, but also due to failure of physics obeying-ness (e.g. feet skating, floating, penetrating a surface), which is not explicitly enforced by deep generative models.

7.2. Physics Simulator as Self-Correction Function

For our self-correction function, we use Universal Humanoid Control (UHC) (Luo et al., 2021), which is an imitation policy that operates inside the MuJoCo physics simulator (Todorov et al., 2012). Given an input sequence of humanoid skeleton poses, UHC attempts to imitate the motion sequence, constrained by the laws of physics imposed by the physics simulator, and it outputs a new motion sequence that is the closest possible approximation it can replace it with. For example, if an input motion sequence violates the laws of physics by having a foot penetrate through the floor, then the motion sequence output by UHC will attempt to remove that physically impossible artifact while maintaining the semantic integrity of the original input motion. We use VPoser (Pavlakos et al., 2019) and SMPL (Loper et al., 2015) to translate joint representations between the human motion generator and the physics simulator.

The physics simulator allows us to self-correct a synthesized motion automatically. Our underlying assumption is that by enforcing the physics obeying-ness (via the simulator) and closeness to the synthesized motion (via the imitation objective), the self-correction function would act as similar as an idealized corrector as possible.

7.3. Experimental setup

We preprocess the MoVi (Ghorbani et al., 2021) subset of HumanML3D (Guo et al., 2022) using the official code implementation of HumanML3D. We filter out movements involving interactions with chairs, as UHC by default does not handle human-object interactions. We take as our train split the train split from HumanML3D, intersected with our filtered subset of MoVi, and likewise for the test split. This procedure yields a train set of size $n = 2794$ and a test set of size 546. We further randomly select a smaller training set of $n \in \{64, 128, 256\}$ examples, to simulate the more challenging scenario when the initial generative model is sub-optimal (due to data scarcity). The smaller data also enables us to explore larger synthetic augmentation percentage due to compute constraints. From here, the iterative re-training procedure follows Algorithm 1. We spell it out in this concrete experimental setup.

We first train on the ground truth train split until the model is nearly converged, using all the default hyperparameters from MDM. We evaluate and save this last checkpoint from generation 0. From here, for each generation $t \in \{1, 2, \ldots, 50\}$, we run three sets of experiments.

A. *Baseline*: fine-tune the latest checkpoint from generation $t-1$ for m batches on ground truth dataset $\mathcal{D}_{\text{real}}$.

B. *Iterative fine-tuning:* fine-tune the latest checkpoint from generation $t-1$ on $\mathcal{D}_{\text{real}} \cup \mathcal{D}_{\text{synth},t-1}$ for m batches. Here, $\mathcal{D}_{\text{synth},t-1}$ is a synthetic dataset of size $\lfloor \lambda \cdot n \rfloor$ generated from the checkpoint for generation $t-1$, using randomly chosen prompts from the train split.

C. *Iterative fine-tuning with self-correction:* fine-tune the latest checkpoint from generation $t-1$ on $\mathcal{D}_{\text{real}} \cup \text{UHC}(\mathcal{D}_{\text{synth},t-1})$ for m batches. Here, $\text{UHC}(\mathcal{D}_{\text{synth},t-1})$ denotes a synthetic dataset of size $\lfloor \lambda \cdot n \rfloor$ generated from the latest checkpoint for generation $t-1$, using randomly chosen prompts from the train split, which is then corrected by UHC.

We experiment with synthetic augmentation percentages $\lambda \in \{0.05, 0.10, 0.15, 0.20, 0.25\}$ on the larger dataset; we set the number of batches seen during generation 0 to be 3125, and the number of batches seen for each later generation to be $m = 625$. Separately, we experiment with synthetic augmentation percentages $\lambda \in \{0.25, 0.50, 0.75, 1.00\}$ on the smaller datasets; we set the number of batches seen during generation 0 to be $78 * k$ for

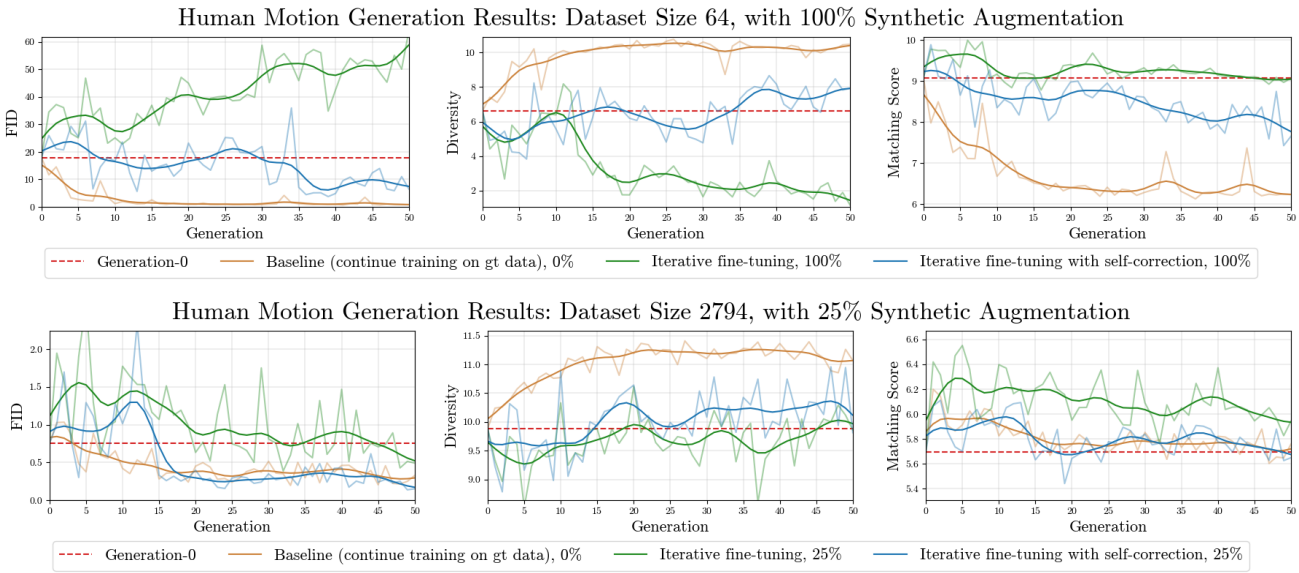

Figure 4. Results from our human motion experiments on iterative fine-tuning with self-correction. These graphs show evaluation metrics for the last checkpoint for every generation. This is the checkpoint used for sampling in the iterative fine-tuning experiments, and it is also the checkpoint where training is resumed with this new partially synthesized dataset. We can see that **with self-correction**, the iterative fine-tuning procedure more stably converges to a better FID score, and more quickly. When the dataset size is smaller ($n = 64$, above) we can see that iterative fine-tuning **with no self-correction** has a flat Matching score, as well as diverging FID and Diversity scores, indicating model collapse. And when the dataset size is larger ($n = 2794$, below), there is less collapse for iterative fine-tuning with no self-correction, although the variance of the FID score is worse, as is the average FID across generations. In both cases, we see that iterative fine-tuning **with self-correction** outperforms iterative fine-tuning **with no self-correction**, and is competitive with the **baseline** after many generations.

dataset size $64 * k$, and the number of batches seen for each later generation $t > 0$ to be $m = 16$. We choose to control how many data points the model sees across each generation, rather than controlling some other quantity like the number of epochs, as this allows each experiment to compare against its baseline in a controlled way, which in turn allows them to compare against each other in a controlled way.

We compute every evaluation one time for each checkpoint using the evaluation script provided in the original MDM codebase. Regardless of the train split size, we perform sampling for evaluation using all 546 motion sequences from the test split, since the FID score is sensitive to generated dataset size. We use the same hyperparameters as those used for MDM, including batch size 64, AdamW (Loshchilov & Hutter, 2019) with learning rate $1e-4$, and classifier-free guidance parameter 2.5. And for UHC we used the `uhc_explicit` model for imitation.

7.4. Quantitative Analysis of Results

For each of these experiments we report the metrics from MDM, as used by (Guo et al., 2022): FID measures how similar the distribution of generated motions is to the ground truth distribution; Diversity measures the variance of the generated motions; and Matching Score measure how well the generated motions embody the given text prompt. In Figure 4 we present results from experiments on our 64-size dataset with 100% synthetic augmentation, as well as our 2794-size dataset with 25% synthetic augmentation.

Our experimental results confirm our theoretical results, that iterative fine-tuning with self-correction outperforms iterative fine-tuning without self-correction, in the sense that the graphs are generally more stable across generations, and approach better evaluation metric values. In particular, Theorem 4.3 and Corollary 4.5 claim that any amount of idealized self-correction will improve the stability bound during iterative fine-tuning. Our results in Figure 4 demonstrate that the FID score is lower and more stable across generations when applying self-correction, and generally higher and less stable than the baseline, where there is no self-consuming training at all. We conduct experiments across multiple seeds, and we find empirically that this general phenomenon holds consistently, where the self-correction technique consistently yields improved training dynamics over iterative fine-tuning with no correction. Graphs from these runs can be found in Appendix G.

Our experimental results also provide empirical evidence for Conjecture 4.7. Observe that in the baseline experiments in Figure 4, the FID score decreases across generations, which indicates that the initial model parameters θ_0^n are not that close to the optimal model parameters $\theta^\star$; additionally, the augmentation percentages considered in the graph

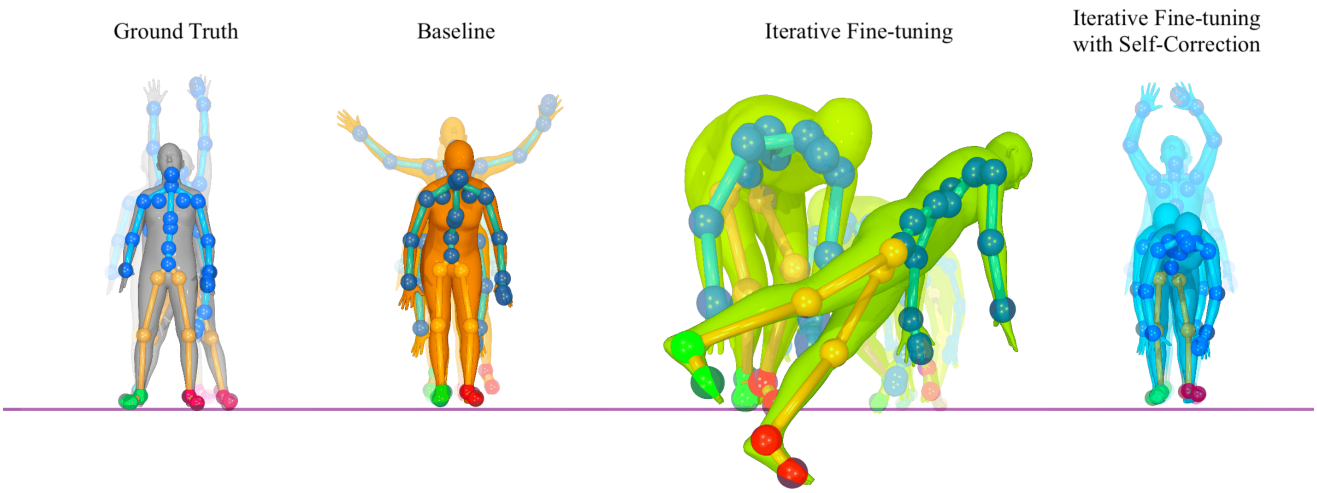

Figure 5. How does the self-correction operation affect iterative fine-tuning, qualitatively? Here we present some visualizations. The prompt which describes the ground truth motion, and which we use to generate the three other motions, is: *"a person stands with feet wide, stretches both hands up over his head and then swings down by the waist and hangs arms down before standing up"*. We can see that the **iterative fine-tuning model** produces a motion where the human moves closer to the camera than the others; this is evidence of model collapse, as moving feet is irrelevant to the prompt. Additionally, this motion produces single frames that suddenly snap to a physically impossible position–note the leg penetration through the ground plane. These negative artifacts do not exist in the motions synthesized from the **ground truth**, **baseline model**, or **iterative fine-tuning with self-correction model**. Lastly, we note that the iterative fine-tuning motion depicted here is semantically similar to crawling. We observe in our experiments with smaller dataset sizes that the iterative fine-tuning model generates less diverse outputs than the baseline model and the iterative fine-tuning with self-correction model, and that this crawling pattern appears more often in the latter. Each snapshot is taken at exactly frame 105 of their respective videos. The two motions on the right come from models that were iteratively fine-tuned for 50 generations, with a train set of size $n = 64$, and a synthetic augmentation percentage of 25%. For all pictures of the human, the camera is fixed at the same position, and for consistency the image is not resized.

are 25% and 100%. Conjecture 4.7 claims that performing self-correction during iterative fine-tuning improves performance, even when the initial model weights are sub-optimal and simultaneously the synthetic augmentation percentage is large. This claim is confirmed by Figure 4. We direct the curious reader to Appendix F, where we present graphs for all of the above listed training set sizes and augmentation percentages, providing additional empirical evidence for Theorem 4.3, Corollary 4.5, and Conjecture 4.7.

7.5. Qualitative Analysis of Results

We visually inspect the generated human motion sequences in order to analyze what concrete effect the self-correction has on iterative fine-tuning. We find that the correctness and diversity of synthesized motions are improved by the self-correction procedure, in agreement with our quantitative analysis in Subsection 7.4. We present snapshots of our synthesized motions in Figure 5, and we analyze the motions in the caption. In short, we find that physics-disobeying artifacts such as floor penetration or floating become more pronounced without the self-correction. We also find that in the model without self-correction, the humanoid sometimes performs movements completely unrelated to the prompt; our model with self-correction fixes these negative phenomena. We direct the curious reader to Appendix E, where we present more examples from our qualitative analysis, as well as our project webpage, where we provide side-by-side video comparisons.

8. Conclusion

Our paper investigates the learning of generative models when the training data includes machine-generated contents. We investigate how self-correction functions, which automatically correct synthesized data points to be more likely under the true data distribution, can stabilize self-consuming generative model training. Our theoretical results show that self-correction leads to exponentially more stable model training and smaller variance, which we illustrate with a Gaussian toy example. We then demonstrate how physics simulators can serve as a self-correction function for the challenging human motion synthesis task, where models trained with our self-correcting self-consuming loops generate higher quality motions, and manage to avoid collapse even at a high synthetic data to real data ratio. Future work includes exploring self-correcting functions for more diverse applications, such as language modeling and text-to-image generation, and investigating when self-consuming training may lead to overall better generative models.